

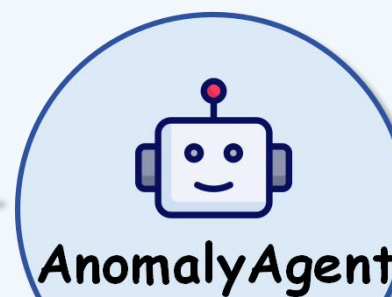


Normal Image (I)

Evaluate and edit the provided image (Category: **hazelnut**) to synthesize a high-quality and physically realistic **hole** anomaly.

Instruction (c, a)

- Prompt Generation
- Knowledge Retrieval
- Mask Generation
- Quality Evaluation
- Image Generation



AnomalyAgent

Adaptive Tool Call



Environment Feedback



Environment

CoT 1



Prompt Generation

`` Change only [**area**] to introduce [**anomaly**]. Keep the rest unchanged. ``

CoT 2



Image Generation



Edit

CoT 3



Quality Evaluation



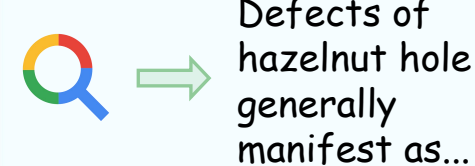
score: 2

review: ...unrealistic...

CoT 4



Knowledge Retrieval



Defects of hazelnut hole generally manifest as...

CoT 5



Image Generation

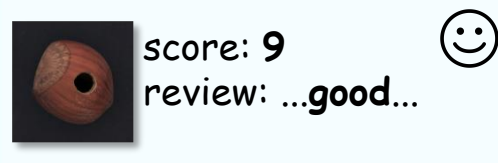


Edit

CoT 6



Quality Evaluation



score: 9

review: ...good...

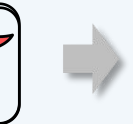
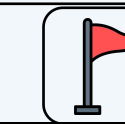
CoT 7



Mask Generation



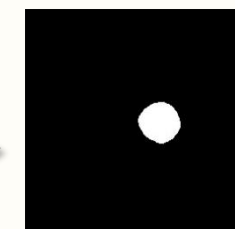
CoT 8



End of the Task

Environment Feedback

Output



Mask (M)



Anomaly (I')

SFT Training

Synthetic Trajectory

RL Training

$$\alpha R_{task} + \beta R_{ref} + \gamma R_{beh}$$